CLEAN DEVELOPMENT MECHANISM
PROJECT DESIGN DOCUMENT FORM (CDM-PDD)
Version 03 - in effect as of: 28 July 2006

CONTENTS

Annexes

SECTION A. General description of project activity

A.1 Title of the project activity:

Inner Mongolia Wudaogou 50.25MW Wind Power Project
Version: 3.0
Date: 14/11/2007.

A.2. Description of the project activity:

Inner Mongolia Wudaogou 50.25MW Wind Power Project (hereafter referred as the proposed project) is a grid connected renewable energy project located in Inner Mongolia Autonomous Region, Northeast China. The objective of the proposed project is to generate electricity using state-of-the-art wind power generation technology and to sell into the State Power Grid. The proposed project will achieve CO_2 emission reduction by replacing electricity generated by fossil fuel fired power plant connected into China Northeast Power Grid.

The proposed project is located in Wengniute County, Inner Mongolia Autonomous Region. Totally 67 wind turbines with a nominal capacity of 750KW have been installed, providing a total capacity of 50.25MW. The area has relative rich wind resource. With an average annual generation of 124.66GWh, the proposed project is estimated to produce 142848 tonnes CO_2 emission reduction annually.

The wind power generation technology adopted in the proposed project is zero emission generation technology. The electricity generated by the proposed project will replace electricity generated by fossil fuels, therefore, the proposed project could reduce the fossil fuel consumptions in connected power grid and as a result reduce CO_2 emissions. However, due to the facts listed as follows, without CDM related income, the proposed project would not be implemented, as a result, the emission reductions would not occur:

- ♦ Comparing with normal fossil fuel fired power plant, the wind power project has much high capital cost per kW and less annual operation hours. The electricity sale price for the proposed project will be 0.5073 RMB /kWh (Excl. VAT), and the IRR of the proposed project is only 7.07%. This is just a price expected, not decided eventually. Seen from the wind power price policies, it is difficult to reach the price anticipated. Therefore, without CDM related income, the proposed project is lack of economic competitiveness.

- ♦ Although China has enacted “Renewable Energy Law” to promote the development of renewable energies including wind power, the specific implementation rules are not yet established.

Being as an environmentally sound energy supply technology, wind power is a priority development project in China. The contributions of the proposed project to sustainable development goal are summarized as follows:

♦ Being located in a power grid dominated by thermal power plants, development of the proposed project will not only reduce GHG emissions but also mitigate local environmental pollution caused by air emissions from thermal power plants.
♦ The proposed project could be helpful to diversify power mix of Northeast Power Grid.
♦ Chinese government has established policies to encourage investment in Northeast China in order to accelerate local economic development. The proposed project could contribute to meeting local electricity

demand, therefore undoubtedly boosts the economy in the local region.

- ♦ Reducing the dependence on exhaustible fossil fuels for power generation;
- ♦ Creation of employment.

A.3. Project participants:

Name of Party involved (*) ((host) indicates a host Party)	Private and/or public entity(ies) project participants (*) (as applicable)	Kindly indicate if the Party involved wishes to be considered as project participant (Yes/No)
China (host)	Chifeng Xinsheng Wind Power Co. Ltd	No
United Kingdom	EDF Trading Limited	No

A.4. Technical description of the project activity:

A.4.1. Location of the project activity:

A.4.1.1. Host Party(ies):

China

A.4.1.2. Region/State/Province etc.:

Inner Mongolia

A.4.1.3. City/Town/Community etc:

Wengniute County

A.4.1.4. Detail of physical location, including information allowing the unique identification of this project activity (maximum one page):

The proposed project is located in Wengniute couty, East Inner Mongolia, its geographical coordinates are north latitude 42°39′-42°42′and east longitude 117°54′-118°01′, and its altitude is around 1700m. It is 90km away from Wudan town , where Wengniute county government is located . The detailed location of the proposed project is shown in Figure 1.

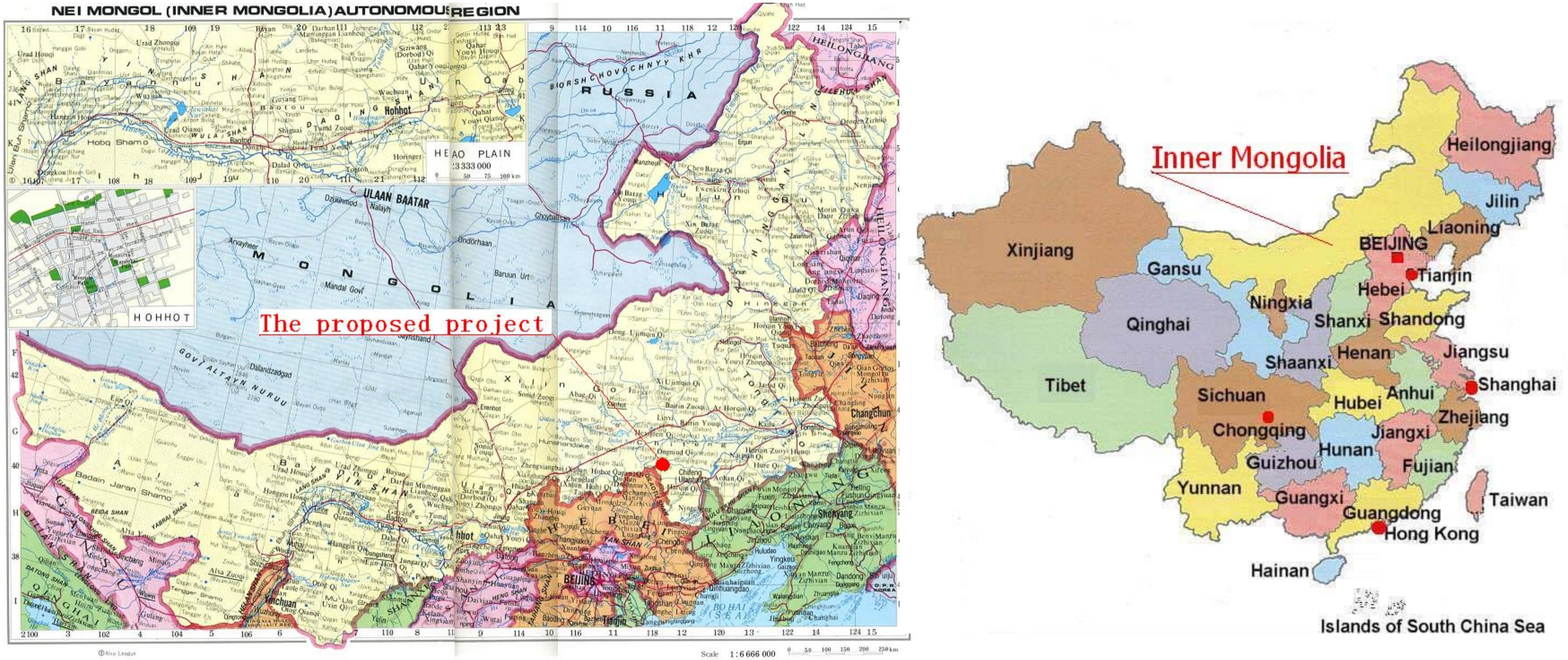

Figure 1 Location of the proposed project

A.4.2. Category(ies) of project activity:

This category would fall within sectoral scope 1: energy industries.

A.4.3. Technology to be employed by the project activity:

Totally 67 wind turbines with a nominal capacity of 750 kW will be installed with 50m as the turbine installation height, providing a total capacity of 50.25 MW. Xinjiang Gold Wind 750kW (S48/750) wind turbines are adopted in the proposed project. Each turbine will have a 690V-to-35kV transformer, from which a 35kV line will link into the on-site 220kV switchgear at the substations established in the proposed project site. By the 220 kV line, the electricity generated by the proposed project are delivered to the power grid.

The wind turbines and transmission facility could be monitored and controlled either by onsite central control room or remotely through Internet.

The wind turbine finally adopted by the proposed project is Xinjiang Goldwind 750kW (S48/750) produced in China. The designs of Goldwind S48/750kW wind turbine, is horizontal axis, three-blade, upwind, stall regulation, asynchronous generator, which is mature technology widely adopted in China. The development of the proposed project will contribute to promoting application of such type of wind turbine, accelerating the accumulation of experiences and absorption of this kind of technology and advancement of domestic wind power technology.

A.4.4 Estimated amount of emission reductions over the chosen crediting period:

A renewable crediting period is selected for the proposed project activity. A reduction of approximately 999936tCO_2e is forecast for the first 7-year crediting period in the table below.

Years	**Annual estimation of emission reductions in tonnes of CO_2 e**
2008.3-2008.12	119040
2009	142848
2010	142848
2011	142848
2012	142848
2013	142848
2014	142848
2015.1-2015.2	23808
Total estimated reductions (tonnes of CO_2e)	999936
Total number of crediting years	7
Annual average over the crediting period of estimated reductions (tonnes of CO_2e)	142848

A.4.5. Public funding of the project activity:

No public funds from Annex I countries is involved in the proposed project.

SECTION B. Application of a baseline and monitoring methodology

B.1. Title and reference of the approved baseline and monitoring methodology applied to the project activity:

Baseline methodology:

ACM0002(Version 6): Consolidated baseline methodology for grid-connected electricity generation from renewable sources.

"Tool for the Demonstration and Assessment of Additionality (version 03)".

Monitoring methodology:

Approved consolidated monitoring methodology ACM0002 (Version 6): "Consolidated monitoring methodology for zero-emissions grid-connected electricity generation from renewable sources".

Reference: UNFCCC website: http://cdm.unfccc.int/methodologies/PAmethodologies/approved.html

B.2. Justification of the choice of the methodology and why it is applicable to the project activity:

The proposed project can meet the applicability criteria of the baseline methodology (ACM0002), therefore, the methodology is applicable to the proposed project.

- The proposed project is a grid-connected zero-emission renewable power generation activity from wind source;
- The proposed project is not an activity that involves switching from fossil fuels to renewable energy at the proposed project site.
- The power grid (the Northeast Power Grid) which the proposed project is to be connected to is clearly identified and information on the characteristics of this grid is publicly available.
- The methodology will be used in conjunction with the approved consolidated monitoring methodology ACM0002 (Consolidated monitoring methodology for grid-connected electricity generation from renewable sources).

The latest version of ACM0002 (version 6, effective from 19 May 2006) has been applied.

B.3. Description of how the sources and gases included in the project boundary:

According to the methodology ACM0002, since the proposed project is a grid connected wind power project, only CO_2 emission from fossil fuels fired power plants in baseline scenario need to be considered.

	Source	Gas	Included?	Justification / Explanation
	Power plants	CO2	Yes	Major emission sources

	connected to the Northeast Power Grid	CH4	No	Excluded for simplification. This is conservative.
		N2O	No	Excluded for simplification. This is conservative.
Project Activity	On-site fuel combustion to the project activity	CO2	No	No emission
		CH4	No	No emission
		N2O	No	No emission

According to the methodology (ACM0002) (version 6), the relevant grid definition should be based on the following considerations:
(a) Use the delineation of grid boundaries as provided by the DNA of the host country if available; or
(b) Use, Where DNA guidance is not available, the following definition of boundary:
- In large countries with layered dispatch systems (e.g. state/provincial/regional/national) the regional grid definition should be used. A state/provincial grid definition may indeed in many cases be too narrow given significant electricity trade among states/provinces that might be affected, directly or indirectly, by a CDM project activity;
- In other countries, the national (or other largest) grid definition should be used by default.

According to above requirements, a project electricity system is defined by the spatial extent of the power plants that can be dispatched without significant transmission constraints. The proposed project delivers its electricity generated to Northeast Power Grid. Therefore, the project's system boundaries are defined as.the project site and the Northeast Power Grid.

The Northeast Power Grid consists of the Heilongjiang Power Grid, Jilin Power Grid, Liaoning Power Grid and East Inner Mongolia Grid. However, the data in the China Energy Statistical Yearbooks and the China Electric Power Yearbooks, the primary data sources used in the calculation of the combined margin emission factor (see below), are presented by province instead of by power grid. This means that no recent data are available on the fuel consumption of the power plants connected to the East Inner Mongolia Grid. Thus it is not possible to separate the data from the West Inner Mongolia Grid and the East Inner Mongolia Grid from the Inner Mongolia Grid. To deal with this problem, we have used data for the other three provinces (Liaoning, Jilin, Heilongjiang) to calculate the combined margin emission factor.[1] It is underlies the calculation of the emission factors published by NDRC on 16 October 2006.

B.4. Description of how the baseline scenario is identified and description of the identified baseline scenario:

Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Project Document Design.

[1] It is conservative to neglect the East Inner Mongolia data. The combined margin emission factor of the East Inner Mongolia Grid is much larger than that of the other three province grids, and from the latest available data from the *China Electric Power Yearbook* 2003 page 593, in 2002, for the East Inner Mongolia, percentage of the thermal plants generation in the total generation is 99.89%, larger than the percentage of the other three provinces, 94.56%. Additionally, in 2002, the Eastern Inner Mongolia Grid makes up only 10.72% generation of the overall Northeast China Grid.

To provide the same output or services comparable with the proposed CDM project activity, these alternatives are to include:

a) The thermal power plant with the same capacity or the same annual electricity output as the proposed project.
b) The proposed project not undertaken as a CDM project activity but as a commercial project.
c) The other renewable energy power plant with the same capacity or the same annual electricity output as the proposed project.
d) The Northeast Power Grid as the provider for the same capacity and electricity output as the proposed project.

The alternative b) is unrealistic and should be eliminated from the following consideration because the investment analysis in Step 2 of section B.5 will show the proposed project not undertaken as a CDM project and without CERs income is lack of the attraction for the potential investors.

The alternative c) is also unrealistic and should be eliminated from the following consideration because the analysis in Step 1a of section B.5. will show that the proposed project owner is only dedicated to wind power development in Inner Mongolia according to the business scope, and has no experience and ability to develop other renewable energy power plants.

The alternative a) is also unrealistic and should be eliminated from the following consideration because the analysis in Step 1b of section B.5. will show that the thermal power plant with the same annual electricity output as the proposed project does not comply with Chinese legal and regulatory requirement.

In conclusion, the alternative d) is the only realistic, credible alternative which is in compliance with all applicable legal and regulations, which is considered as the baseline scenario of the proposed project.

B.5. Description of how the anthropogenic emissions of GHG by sources are reduced below those that would have occurred in the absence of the registered CDM project activity (assessment and demonstration of additionality):

CDM has been considered seriously before the starting date of the project activity, which is evidenced by a loan permission letter from China Development Bank on 9 March 2006. The evidence is available for DOE.

The loan permission letter clearly states that: Considering that the project owner will apply for CDM, the proposed project can, in addition to normal operation revenues, get CDM income through the transfer of greenhouse gas emission reductions, the repayment ability of the project owner is increased, the China Development Bank permits the loan to the project owner.

The following steps are used to demonstrate the additionality of the proposed project according to “Tools for the demonstration and assessment of additionality (version 03)” agreed by Executive Board and requested by the baseline methodology (ACM0002).

Step1. Identification of alternatives to the project activity consistent with current laws and regulations.

The objective of this step is to identify realistic and credible alternatives to the proposed project that can be the baseline scenario through the following sub-steps:

Sub-step 1a. Define alternatives to the project activity.

To provide the same output or services comparable with the proposed CDM project activity, these alternatives are to include:

a) The thermal power plant with the same capacity or the same annual electricity output as the proposed project.
b) The proposed project not undertaken as a CDM project activity but as a commercial project.
c) The other renewable energy power plant with the same capacity or the same annual electricity output as the proposed project.
d) The Northeast Power Grid as the provider for the same capacity and electricity output as the proposed project.

The alternative b) is unrealistic and should be eliminated from the following consideration because the investment analysis in Step 2 will show the proposed project not undertaken as a CDM project and without CERs income is lack of the attraction for the potential investors.

The alternative c) is also unrealistic and should be eliminated from the following consideration. Besides wind energy, solar PV, geothermal, biomass and hydro are the possible grid-connected renewable energy technologies that could be applied in the Northeast Power Grid. However, the proposed project owner is only dedicated to wind power development in Inner Mongolia according to the business scope, and has no experience and ability to develop other renewable energy power plants. Hence, the alternative c) is unrealistic too.

To summarize, the realistic and creditable alternatives that can provide the same output or services as the proposed project are a) and d).

Step1b. Consistency with mandatory laws and regulations.

The applicable legal and regulatory requirement for the proposed project include laws, central government regulations, local regulations, departmental rules and disciplines related to electricity and environment protection.

The related laws and regulations can be found and downloaded on the website of State Electricity Regulatory Commission (SERC) and National Development and Reform Commission (NDRC): http://www.serc.gov.cn/opencms/export/serc/laws/index.html and http://nyj.ndrc.gov.cn .

According to the applicable laws and regulations, the alternative a) should be eliminated from the following consideration because it does not comply with the national regulation for controlling small scale thermal power plant. To provide the same output as the proposed project, the alternative thermal power plant will has the capacity less than 50 MW then will be categorized as the small scale thermal power plant and should be shut down according to the regulations from NDRC. According to this regulation, the thermal power plant under 50 MW should be shut down and the construction of coal power plant under 135 MW will be forbidden within the connected area.

In conclusion, the alternative d) is the only realistic, credible alternative which is in compliance with all applicable legal and regulations, which is considered as the baseline scenario of the proposed project.

Step2. Investment analysis.

This step will determine whether the proposed project is the economically or financially less attractive than other alternatives without the revenue from the sale of CERs.

Sub-step 2a. Determine appropriate analysis method.

Three options can be applied for the investment analysis: the simple cost analysis, the investment comparison analysis and the benchmark analysis.

The simple cost analysis is not applicable for the proposed project because the project activity will produce economic benefit (from electricity sale) other than CDM related income. The investment comparison analysis is also not applicable for the proposed project because the baseline scenario, providing the same capacity or electricity output by the Northeast Power Grid, is not a project.

To conclude, the benchmark analysis will be used to identify whether the financial indicators (such as IRR or NPV) of the proposed project is better than relevant benchmark value.

Sub-step 2b Apply benchmark analysis.

In accordance with *Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects* issued by former State Power Corporation of China, there is not yet such a financial internal rate of return (FIRR) as benchmark in China's power generation industry to date. However, the project FIRR shall not be lower than 8 percent considering economic assessments of hydropower projects, fossil fuel fired projects, transmission and substation projects, especially the interest rate of commercial loans over five years[2]. Nowadays many of China's existing wind power projects have applied it as the benchmark IRR.

The FIRR of the proposed project is calculated and compared as follows.

Sub-step 2c. Calculation and comparison of financial indicators.

The parameters needed for calculation of IRR are from the feasibility study report of the proposed project except Total Investment[3].

Table 1 Main parameters for calculation of financial indicators Wudaogou windfarm

Items	Unit	Amount
Capacity	MW	50.25
Total Investment	Million RMB	472.66
Annually output	GWh/year	124.66
Electricity Tariff (Excl. VAT)	RMB/kWh	0.5073
Value Added Tax (VAT)	%	8.5
Income tax	%	15
Expected CERs Price	EUR/tCO_2	10.21

[2] State Power Corporation of China. Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects. Beijing: China Electric Power Press, 2003

[3] The total investment is 453.56 Million RMB in the Feasibility Study Report. But there is an additional investment of 19.1 Million RMB approved by the government on December 9th 2005.

Project life time	Year	20

The financial indicators (FIRR) with and without income from selling CERs are listed in the following table. Without income from selling CERs, the FIRR of the proposed project is lower than the benchmark FIRR and the proposed project is financially unacceptable because of its low profitability. While considering such income, the financial acceptance will be changed, the FIRR of the proposed project is better than the benchmark then the proposed project is financially acceptable.

Table 2 Comparison of financial indicators with and without income from CERs for Wudaogou wind farm

Items	Unit	Without income from CERs	Benchmark	With income from CERs
FIRR	%	7.07	8	9.74

Sub-step 2d. Sensitivity analysis.

The objective of this sub step is to show the conclusion regarding the financial attractiveness is robust to reasonable variations of the critical assumptions.

Four factors are considered in following sensitivity analysis:

1) Total investment.
2) Annual operation and maintenance cost.
3) Tariff.
4) Plant load factor(PLF)

Assuming the above Four factors vary in the range of -10%-+10%, the FIRR of Wudaogou sub-wind farm (without income from selling CERs) varies to different extent, as shown in figure 2.

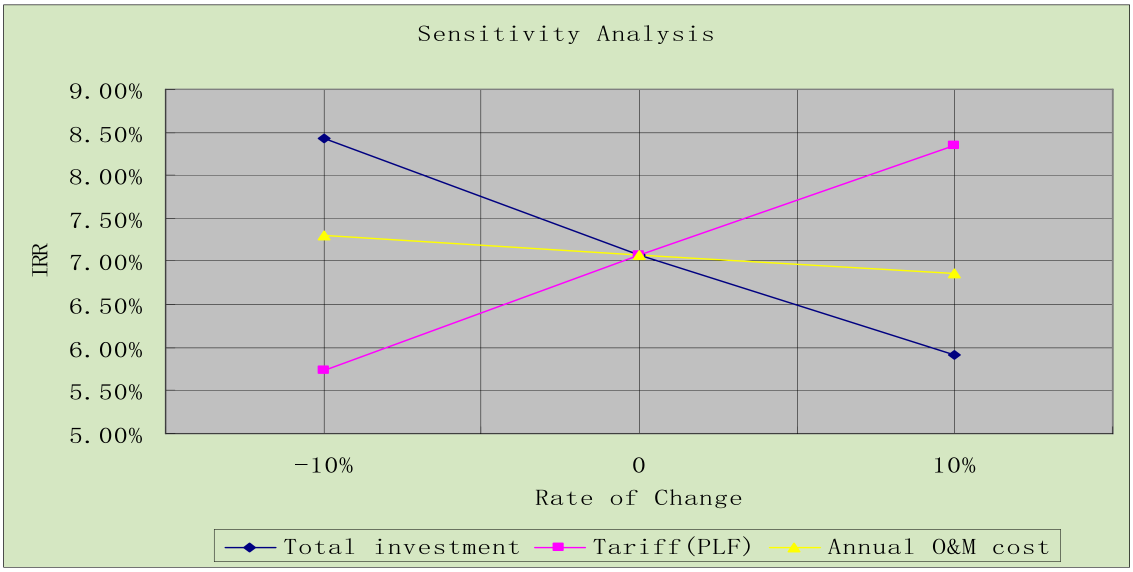

Figure 2 Sensitivity analysis of the proposed project

The change of total investment is an important factor affecting the financial attractiveness of the proposed project. In the case that the total investment decreases by 6.8%, the FIRR of the proposed project begins to exceed the benchmark. But the wind farm construction period is finished, the investment is all commissioned. So the total investment is not likely to decrease by 6.8%. Hence within the reasonable range of total investment, the proposed project is always lack of financial attractiveness.

The next important factor for financial attractiveness is the expected power tariff. In the case that the expected power tariff increases by 7.2%, the FIRR of the proposed project begins to exceed the benchmark, the conclusion of financial additionality begins to be in question. However there is extremely unlikely for the tariff of the proposed project to have an increase of 7.2%. According to China's Management Rules on Tariff issued by NDRC[4], the tariff of the un-tendering projects should be determined by the government with reference to the tariff of tendering wind projects. By this pricing principle, China government is gradually lowering down the wind power in-grid tariff[5]. The highest tariff of the approved tendering wind power project is 0.5190 Yuan/kWh (incl. VAT) before 2006[6], which is lower than the expected tariff in the Feasibility Study Report of the proposed project, so the final approved tariff for the proposed project should be lower than the tariff in the Feasibility Study Report of the proposed project. And, once the tariff is issued, it will be strictly regulated by the government, neither the project owner nor the grid company can change it. Given the above, it is impossible that the actual tariff for the proposed project will be 7.2% higher than the expected tariff 0.5073 Yuan/kwh(Excl. VAT). Therefore within the reasonable range of expected power tariff, the proposed project is always lack of financial attractiveness.

Furthermore, the information of the plant load factor (PLF) is critical since the PLF reflects the achievable annual generation output of the proposed project. It shows the sensitivity around the PLF is similar to the sensitivity around the tariff (both impact the turnover the same way). FIRR of the proposed project would go beyond the benchmark when the PLF is 7.2% higher than the 28.7% assumption. Considering the PLF value is positive correlation with the wind speed, and the PLF value in the feasibility study report is based on the wind speed data of 2005. The data provided by local meteorological station shows that the average wind speed in 2005 is 2.9m/s, which is higher than the average wind speed 2.8m/s of the latest 30 years. The wind speed exceeds 3.1m/s when the wind speed is 7.2% higher than 2.9m/s. However, the average wind speed of the project site tends to decrease and it is 2.8 m/s over the past 30 years (from 1975 to 2004)[7] as shown following table 3. The probability the speed beyond 3.1m/s is very small in the latest 10 years. Therefore, it is very hard that the PLF is 7.2% higher than the 28.7% assumption to the proposed project.

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[4] Interim Regulation for Tariff of Renewable Energy Power Generation and Appointment of Expenses FAGAIJIAGE(2006) No.7

[5] http://www.eri.org.cn/manage/upload/uploadimages/eri200672795944.pdf

[6]http://news.cepee.com/text.php?NewsId=25998

[7] feasibility study report of the proposed project p12

Year	1957	1958	1959	1960	1961	1962	1963	1964	1965	1966	1967	1968
Average wind speed	3.6	3.6	3.5	3.6	3.5	3.9	4.0	3.2	3.2	3.4	3.0	3.1
Year	1969	1970	1971	1972	1973	1974	1975	1976	1977	1978	1979	1980
Average wind speed	3.0	2.9	3.3	3.0	2.8	2.8	2.6	2.7	2.5	2.9	2.6	2.9
Year	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
Average wind speed	3.0	3.2	3.1	3.0	2.7	2.8	3.1	3.4	3.1	2.6	2.7	2.5
Year	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004
Average wind speed	2.8	2.6	2.9	3.0	2.8	2.6	2.7	2.5	2.5	2.5	2.8	2.9

Table 3: The Average Wind speed provided by local meteorological station(m/s)

The impact of the annual O&M cost is the slightest, the FIRR of the proposed project begins to exceed the benchmark when the annual O&M cost decreases by much more than 10%. Since the wind turbines operate in high latitude area, such reduction of O&M cost is lack of possibility for the proposed project, therefore, the proposed project is always lack of financial attractiveness within the reasonable range of annual O&M cost.

To conclude, under the reasonable variations in the critical assumptions, the conclusion regarding the financial additionality is robust and supported by sensitivity analysis.

Step 3. Barrier analysis

Investment analysis has argued that the project is the economically less attractive than other alternatives without the revenue from the sale of CERs. According to "Tool for the Demonstration and Assessment of Additionality (version 3)", this PDD skips the barrier analysis and argues the additionality

Step 4. Common practice analysis

Sub-step 4a. Analyze other activities similar to the proposed project activity:

Table 4 Wind Farms in Inner Mongolia

Project Title	**Commissioning Date**	**Capacity (MW)**	**Note**
Saihanba(North) Wind Farm	Under construction	45.05	Registered CDM project
Saihanba(East) Wind Farm	Under construction	45.05	Registered CDM project
Inner Mongolia Chifeng Dongshan wind Farm	Under construction	49.3	Registered CDM project

Dali wind Farm	1999-2003.12	39.75	Demonstration project
Datang Huitengliang Wind Farm	Under construction	49.50	Registered CDM project
Guohua Huitengliang Wind Farm	Under construction	48.75	Applying for being a CDM project
Mangniuhai	Under construction	49.3	Applying for CDM project

Source: Shi Pengfei, Statistics of Chinese Wind Energy Installed Capacity in 2005, China Wind Energy, No.1 2006;
http://www.chifeng.cn/article/ReadNews.asp?NewsID=4141&BigClassID=1&SmallClassID=2&SpecialD=0;
http://cdm.ccchina.gov.cn/web/Main.asp?ColumnId=1&ScrollAction=2, 2006-7-31
http://cdm.unfccc.int/Projects/index.html;
http://wais.ee.kuas.edu.tw/energyworld/specialtopics/Wind/ref/ref/%AE%D7%A8%D2%A4%C0%AAR.htm

Sub-step 4b. Discuss any similar options that are occurring:

The existing wind farm projects do not call into question the claim that the proposed project is financially unattractive as discussed in Step 2.

Dali Wind Farm is a demonstration project and therefore enjoyed favourable treatments which are not be available for the proposed project. Dali Wind Farm was a demonstration project and therefore enjoyed favourable treatments which are not be available for the proposed project. During the first phase of Dali wind farm, the capital 3,700,000 US dollar for purchasing 7 Denmark 750 kilowatt wind turbines was from the loan of Dutch Government. The second phase of Dali wind farm was a demonstration project and enjoyed a higher tariff of 0.76 yuan/kwh[8]. After the price cap regulation for wind power electricity tariff, such high tariff is impossible for wind farm developers. The third phase of Dali wind farm was the fourth issue of national debt special fund project.

Moreover, Some windfarm CDM projects have been registered by EB in the last few years, and others are applying for CDM registration, because they are financially unattractive without the additional income from CER sales, and face barriers.

To conclude, The existence of these projects in Table 3 does not contradict the claim that the proposed project activity is financially unattractive.

As described above, the proposed project activity passed all criteria of "Tool for the demonstration and assessment of additionality"(version 03). In conclusion, the proposed project is additional and not the baseline scenario.

B.6. Emission reductions:

B.6.1. Explanation of methodological choices:

[8] http://www.fsou.net.cn/html/text/lar/169071/16907181.html

The calculation of the GHG emission reductions by the proposed project is followed the baseline methodology ACM0002 (Ver06). The project electricity system of the proposed project is defined as the Northeast Power Grid (excluding East Inner Mongolia).

Firstly ex ante emission factors of Operating Margin ($EF_{OM,y}$) and Build Margin ($EF_{BM,y}$) were calculated based on the history data of the Northeast Power Grid, which include the installed capacity, electricity generation and different types of fuel consumptions of all the power plants connected into the Northeast Power Grid. Secondly, the baseline emission factor (EF_y) was calculated as a combined margin(CM) of the Operating Margin (OM) and Build Margin (BM) emission factors as described in following three steps. All the calculation is in compliance with requirement of the baseline methodology (ACM0002), the details is listed in the following steps.

Step 1: Calculation the Operating Margin emission factor ($EF_{OM,y}$)

Calculation of OM emission factor should be based on one of the four following methods:

(a) Simple OM, or
(b) Simple adjusted OM, or
(c) Dispatch Data Analysis OM, or
(d) Average OM.

The justifications of the choice of methodology to calculate OM emission factor are as follows.

Method (c): If the dispatch data is available, method (c) should be the first methodological choice. This method requires the dispatch order of each power plant and the dispatched electricity generation of all the power plants in the power grid during every operation hour period. Since the dispatch data, power plants operation data are considered as commercial secret materials and only for internal usage not available publicly. Thus, method (c) is not applicable for the proposed project.

Method (b): Method (b) requires the annual load duration curve of the power grid and the load data of every hour data during the whole year on the basis of the time order. As mentioned above, the dispatch data and detailed load curve data were not available publicly. Therefore, method (b) is not applicable for the proposed project as well.

Method (d): Method (d) will only be used when (1) low-cost/must run resources constitute more than 50% of total grid generation and detailed data to apply option (b) is not available, and (2) where detailed data to apply option (c) above is unavailable. From 2001 to 2005, low-cost/ must run resources in the Northeast Power Grid account for about 4.72%-8.27%. Hence method (d) is not applicable for the proposed project.

Method (a): Method (a) can only be used where low-cost/must run resources constitute less than 50% of total grid generation in: (1) average of the five most recent years, or (2) based on long-term normals for hydroelectricity production. Low operating cost and must run resources typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation. If coal is obviously used as must-run, it should also be included in this list, i.e. excluded from the set of plants. From 2001 to 2005, low-cost/ must run resources in the Northeast Power Grid account for about 4.72%-8.27% (China Electric Yearbooks 2002-2006), the sum of which is much less than 50%. Therefore, method (a) is applicable for the proposed project.
Table 4 Annual electricity generation of Northeast Grid in 2000-2004

No.	Year	Electricity generation （GWh）			low-cost/ must run resources
		Total generation	Fuel-fired power	low-cost/ must run resources	
1	2001	159,748.83	149,655.05	10,093.78	6.32%
2	2002	167,628.39	159,474.93	8,153.46	4.86%
3	2003	165,817.0	157,983.0	7,834.0	4.72%
4	2004	183,090.0	171,267.0	11,823.0	6.46%
5	2005	192,963.0	176,991.0	15,972.0	8.27%

Data source: China Electric Power Yearbook 2002 p. 625
China Electric Power Yearbook 2003 p. 593
China Electric Power Yearbook 2004 p. 709
China Electric Power Yearbook 2005 p. 474
China Electric Power Yearbook 2006 p. 568

In conclusion, method (a) is the only reasonable and feasible method among the four methods for calculating the Operating Margin emission factor ($EF_{OM,y}$) of the proposed project.

According to the ACM0002, the Simple OM emission factor ($EF_{OM,simple,y}$) is calculated as the generation-weighted average emissions per electricity unit (tCO_2/MWh) of all generating sources serving the system, not including low-operating cost and must-run power plants, the detailed formulas are as following:

$$EF_{OM,simple,y} = \frac{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}}{\sum_{j} GEN_{j,y}} \qquad (1)$$

where:

$F_{i,j,y}$ is the amount of fuel i (in a mass or volume unit) consumed by relevant power sources j in year(s) y,

j refers to the power sources delivering electricity to the grid, not including low-operating cost and must-run power plants, and including imports to the grid[9],

$COEF_{i,j,y}$ is the CO_2 emission coefficient of fuel i (tCO_2/ mass or volume unit of the fuel), taking into account the carbon content of the fuels used by relevant power sources j and the percent oxidation of the fuel in year(s) y, and

$GEN_{j,y}$ is the electricity (MWh) delivered to the grid by sources j.

The CO_2 emission coefficient $COEF_i$ is obtained as

$$COEF_i = NCV_i \times EF_{CO_2,i} \times OXID_i \qquad (2)$$

where:

NCV_i is the net calorific value (energy content) per mass or volume unit of a fuel i, (TJ/ mass or volume unit）,

$OXID_i$ is the oxidation factor of the fuel i (IPCC2006 default values),

[9] As described above, an import from a connected electricity system should be considered as one power source j.

$EF_{CO_2,i}$ is the CO_2 emission factor per unit of energy of the fuel i (tCO2e/TJ).

Regarding parameter selection, local values of NCV_i and $EF_{CO_2,i}$ should be used where available. If no such values are available, IPCC 2006 world-wide default values are preferable. The Net Calorific Value (NCV_i) of each type of fossil fuel used in the calculation comes from China Energy Statistic Yearbooks. Emission factors ($EF_{CO_2,i}$)of each type of fossil fuel come from 2006 IPCC default values. For the value of$OXID_i$, it is form 2006 IPCC default values too.

The simple OM emission factor can be calculated using either of the two following data vintages for years(s) y : (1) A 3-year average based on the most recent statistics available at the time of PDD submission, or (2) The year in which project generation occurs, if $EF_{OM,y}$ is updated based on ex post monitoring.

The option (1), a 3-year average based on the most recent statistics available at the time of PDD submission, is adopted for calculating the operation margin emission factor ($EF_{OM,y}$) of the proposed project. The data of installed capacity, electricity generation and fuel consumptions are all from China Energy Statistical Yearbooks and China Electric Power Yearbooks. The net imports from North China Power Grid were all negative for the recent years (China Electric Power Yearbook 2004, 2005, and 2006), thus the value was defined to be zero as requested by the baseline methodology (ACM0002). Based on the calculation results, the Operation Margin emission factor ($EF_{OM,y}$) of Northeast Power Grid is **1.2402 tCO2/MWh.** The detailed data and calculation are listed in the tables of annex 3.

Step2: Calculation the Build Margin emission factor ($EF_{BM,y}$ ***)***

According to the ACM0002, the baseline Build Margin emission factor was calculated using the following formula (3).

$$EF_{BM,y} = \frac{\sum_{i,m} F_{i,m,y} \times COEF_{i,m,y}}{\sum_{m} GEN_{m,y}} \qquad (3)$$

where:

$F_{i,m,y}$ is the amount of fuel *i* (in a mass or volume unit) consumed by *m* power plants in year(s) *y*,

m refers to the power plants included in the sample group determined by the following steps.

$COEF_{i,m,y}$ is the CO_2 emission coefficient of fuel *i* (tCO_2/ mass or volume unit of the fuel), taking into account the carbon content of the fuels used by *m* power plants and the percent oxidation of the fuel in year(s) *y*,

$GEN_{m,y}$ is the electricity (MWh) delivered to the grid by *m* power plants.

According to the baselinle methodology (ACM0002), one of the following two options shall be selected to identify sample group for calculating Build Margin emission factor.

Option 1. Calculate the Build Margin emission factor $EF_{BM,y}$ *ex ante* based on the most recent information available on plants already built for sample *m* at the time of PDD submission. The sample group *m* consists of either

- ♦ The five power plants that have been built most recently, or

- ♦ The power plants capacity additions in the electricity system that comprise 20% of the system generation (in MWh) and that have been built most recently.

Project participants should use from these two options that sample group that comprise the larger annual generation.

Option 2. For the first crediting period, the Build Margin emission factor $EF_{BM,y}$ must be updated annually *ex post* for the year in which actual project generation and associated emission reductions occur. For subsequent crediting periods, $EF_{BM,y}$ should be calculated *ex-ante*, as described in option 1 above. The sample group m consisted of either

- ♦ The five power plants that have been built most recently, or
- ♦ The power plants capacity additions in the electricity system that comprise 20% of the system generation (in MWh) and that have been built most recently.

Project participants should use from these two options that sample group that comprises the larger annual generation.

Power plant capacity additions registered as CDM project activities should be excluded from the sample group m.

For the proposed project, Option 1 was adopted for calculating the Build Margin emission factor.

However, no matter which options mentioned above was adopted for the proposed project, the same issue on data availability must be addressed. Currently, it is very difficulty to get the capacity margin data of power plants in China, since these data as well as generation and fuel consumption data of each power plant are regarded as commercial secrets or only for internal usage. According to the guidance from the CDM Executive Board for a deviation of the baseline methodology of AM0005, which had combined into the baseline methodology of ACM0002, the following deviation was adopted to calculate the Build Margin emission factor.
(http://cdm.unfccc.int/UserManagement/FileStorage/AM_CLAR_QEJWJEF3CFBP1OZAK6V5YXPQK K7WYJ):

1）Use of capacity additions for estimating the build margin emission factor for grid electricity.
2）Use of weights estimated using installed capacity in place of annual electricity generation.
3）Use the efficiency level of the best technology commercially available in the provincial/regional or national grid of China, as a conservative proxy, for each fuel type in estimating the fuel consumption to estimate the build margin (BM).

According to the guidance from EB, the following deviation was adopted to calculate the Build Margin emission factor.

1. Due to breakdown data by power plants are not while the aggregate data by different types of fuels are available, therefore, the *m* sample group will consist of capacity addition by power sources with same fuel instead of by power plants. For the proposed project the *m* sample group will consist of coal fired capacity addition, hydropower capacity addition and other capacity addition;

2. Assuming that all the power plants with same fuel type have equal annual operation hours, the starting year t_0 could be identified which fulfil the following constraint:

$$\sum_i CAP_{i,t-t_0} \geq 20\% \times \sum_i CAP_{i,t} \qquad (4)$$

Where,
t is the recent year of which the latest data is available;
$CAP_{i,t-t0}$ is the capacity addition of type i from year t_0 to year t;
$CAP_{i,t}$ is the installed capacity of type i in year t;

The capacity addition belonging to m sample group thus could be identified. For the proposed project, the most recent year of which data is available is 2005, while t_0=1998, the total capacity addition during 1998 to 2005 accounts for 21.34% of total installed capacity in 2005 (See Annex 3 for detailed calculation)

3. To be conservative, zero emission factors were selected for hydropower capacity and other capacity. The fuel consumption efficiency of 336.66 g standard coal/kWh is selected as the best technology commercially available in China[10], which is used to calculate emission factor of fossil fuel fired capacity addition. It can be acknowledged as the best available data available for estimating the BM in the NEPG.

Step 2a: calculating the respective percentages of CO_2 emissions from coal fired power generation, oil fired power generation, and gas fired power generation against total CO_2 emissions from fossil fuel fired power generation

With the energy balance sheet in China Energy Statistical Yearbook for the most recent year, calculating the respective percentages of CO_2 emissions from coal fired power generation, oil fired power generation, and gas fired power generation against total CO_2 emissions from fossil fuel fired power generation:

$$\lambda_{Coal} = \frac{\sum_{i \in COAL, j} F_{i,j,y} \times COEF_{i,j}}{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}} \qquad (5)$$

$$\lambda_{Oil} = \frac{\sum_{i \in OIL, j} F_{i,j,y} \times COEF_{i,j}}{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}} \qquad (6)$$

$$\lambda_{Gas} = \frac{\sum_{i \in GAS, j} F_{i,j,y} \times COEF_{i,j}}{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}} \qquad (7)$$

where:

$F_{i,j,y}$ is the amount of fuel i (in a mass or volume unit) consumed by province j in year(s) y,

[10] The statistics by State Electricity Regulatory Commission (SERC) on newly built thermal plants in 10th "Five-Year Plan" period. See Annex 3 for detailed information.

$COEF_{i,j,y}$ is the CO_2 emission coefficient of fuel *i* (tCO_2/ mass or volume unit of the fuel), taking into account the carbon content of the fuels consumed by province *j* and the percent oxidation of the fuel in year(s) *y*,

COAL,OIL, and GAS are the aggregation of various kinds of coal, oil, and gas as fossil fuels.

Step 2b: calculating the corresponding emission factor for fossil fuel fired power generation

$$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal,Adv} + \lambda_{Oil} \times EF_{Oil,Adv} + \lambda_{Gas} \times EF_{Gas,Adv} \quad (8)$$

where:

$EF_{Coal,Adv}$, $EF_{Oil,Adv}$ and $EF_{Gas,Adv}$ are the emission factors for the best commercially available technology of coal fired power generation, oil fired power generation, and gas fired power generation, respectively (See Annex 3 for detailed calculation).

Step 2c: Calculating the $EF_{BM,y}$ of local grid

Using the share of different type of capacity in total capacity addition as weight, the weighted average of emission factors of different type capacity is calculated as the Build Margin emission factor $EF_{BM,y}$ of Northeast Power Grid (See Annex 3 for detailed calculation):

$$EF_{BM,y} = \frac{CAP_{Thermal}}{CAP_{Total}} \times EF_{Thermal} \quad (9)$$

where:

CAP_{Total} is the total capacity addition,

$CAP_{Thermal}$ is the fossil fuel fired capacity addition.

Following the three steps above, the Build Margin emission factor $EF_{BM,y}$ of the Northeast China Power Grid is calculated to be: **0.8632 tCO2/MWh.** The detailed calculation and data were listed in the a nnex 3**.**

Data sources for $EF_{OM,y}$ and $EF_{BM,y}$ calculation: Data on installed capacity, power generation, and self-usage rate of power plants are from China Electric Power Yearbooks 1998-2005. The consumption data of various types of fuels and their net caloric values are from China Energy Statistical Yearbooks 2000-2005. The CO_2 emission factors per unit of energy and the oxidation factors are from Revised 2006 IPCC Guidelines for National Greenhouse Gas Inventories.

Step3: Calculation the baseline emission factor (EF_y)

According to the baseline methodology (ACM0002), the baseline emission factor EF_y is calculated as the weighted average of the Operating Margin emission factor ($EF_{OM,y}$) and the Build Margin emission factor ($EF_{BM,y}$),:

$$EF_y = \omega_{OM} \times EF_{OM,y} + \omega_{BM} \times EF_{BM,y} \quad (6)$$

where the weights ω_{OM} and ω_{BM} are 75% and 25% respectively by the default.

The default weights are adopted for the proposed project, the baseline emission factor is:

$EF_y=0.75*EF_{OM,y}+0.25*EF_{BM,y}=$**1.1459 tCO2/MWh.**

B.6.2. Data and parameters that are available at validation:

Data / Parameter:	$F_{i,y}$
Data unit:	t/m^3
Description:	Amount of fuel *i* consumed in year(s) *y* for generation
Source of data used:	China Energy Statistical Yearbook
Value applied:	See Annex 3
Justification of the choice of data or description of measurement methods and procedures actually applied :	Since the detailed fuel consumption data by power plants are not publicly available, therefore the aggregated data by fuel types are used instead.
Any comment:	

Data / Parameter:	$GEN_{i,y}$
Data unit:	MWh
Description:	Electricity (MWh) delivered to the grid excluding low operating cost/must run power plants in year *y*.
Source of data used:	China Electric Power Yearbook
Value applied:	See Annex 3
Justification of the choice of data or description of measurement methods and procedures actually applied :	Since the detailed generation data by power plants are not publicly available, therefore the aggregated data by fuel types are used instead.
Any comment:	

Data / Parameter:	NCV_i
Data unit:	$TJ/t(m^3)$
Description:	Net calorific value (energy content) per mass or volume unit of fuel *i*
Source of data used:	China Energy Statistical Yearbook
Value applied:	See Annex 3
Justification of the choice of data or description of measurement methods and procedures actually applied :	According to ACM0002, the national specific value shall be used preferentially.
Any comment:	

Data / Parameter:	$OXID_i$
Data unit:	%
Description:	Oxidation factor of the fuel *i*
Source of data used:	IPCC2006 default value
Value applied:	See Annex 3

Justification of the choice of data or description of measurement methods and procedures actually applied :	The country specific values of oxidation factors in China are not available. As such IPCC default values are used instead.
Any comment:	

Data / Parameter:	$EF_{CO2,i}$
Data unit:	tCO_2/TJ
Description:	CO_2 emission factor per unit of energy of the fuel *i*
Source of data used:	The emission factor of Carbon comes from IPCC2006 as a default value, $EF_{CO2,I}$ could be calculated by multiplying it by 44 and dividing it by 12.
Value applied:	See Annex 3
Justification of the choice of data or description of measurement methods and procedures actually applied :	The country specific values of fuel CO_2 emission factor in China are not available. As such IPCC default values are used instead.
Any comment:	

Data / Parameter:	Coal fired power supply efficiency
Data unit:	%
Description:	the best commercially available technology of coal fired power generation
Source of data used:	http://cdm.ccchina.gov.cn/web/index.asp
Value applied:	35.82
Justification of the choice of data or description of measurement methods and procedures actually applied :	According to EB guidance, the statistics by State Electricity Regulatory Commission (SERC) on newly built thermal plants in 10th "Five-Year Plan" period can be used.
Any comment:	

Data / Parameter:	Oil and gas fired power supply efficiency
Data unit:	%
Description:	the best commercially available technology of oil and gas fired power generation
Source of data used:	http://cdm.ccchina.gov.cn/web/index.asp
Value applied:	47.67
Justification of the choice of data or description of measurement methods and procedures actually applied :	According to EB guidance, the statistics by State Electricity Regulatory Commission (SERC) on newly built thermal plants in 10th "Five-Year Plan" period can be used.
Any comment:	

B.6.3. Ex-ante calculation of emission reductions:

According to the baseline methodology ACM0002, the GHG emission of the proposed project within the project boundary is zero, i.e. *PEy*=0.

According to the baseline methodology ACM0002, the leakage of the proposed project is not considered, i.e. *Ly*=0.

Therefore, the proposed project activity emissions are zero, i.e. E.1+E.2=*PEy*+*Ly*=0.

According to the descriptions and formulas in section B.6.1, the combined baseline emission factor of the Northeast Power Grid is: ***EFy* =1.1459tCO2MWh.**

According to the Feasibility Study Report of the proposed project, the estimated annual electricity generation delivered to the power grid will be: ***EGy*=124,660MWh.**

The annual emissions of baseline scenario is: ***BEy* =*EGy*×*EFy*=**142848**tCO2.**

The annual emission reductions of the proposed project during the first crediting period are estimated to be:

***ERy* =*BEy*-*PEy* = *BEy* =**142848**tCO2.**

B.6.4. Summary of the ex-ante estimation of emission reductions:

Year	**Estimation of Project activity Emission (tonnes of CO2e)**	**Estimation of baseline emission (tonnes of CO2 e)**	**Estimation of leakage (tonnes of CO2e)**	**Estimation of Emission reductions (tonnes of CO2 e)**
2008.3-2008.12	0	119040	0	119040
2009	0	142848	0	142848
2010	0	142848	0	142848
2011	0	142848	0	142848
2012	0	142848	0	142848
2013	0	142848	0	142848
2014	0	142848	0	142848
2015.1-2015.2		23808		23808
Total (t CO2e)	**0**	**999936**	**0**	**999936**

B.7. Application of the monitoring methodology and description of the monitoring plan:

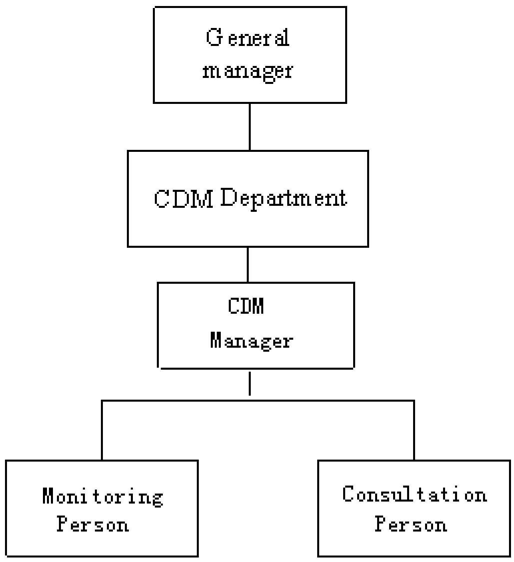
B.7.1. Data and parameters monitored:

Data / Parameter:	**EG_y**

Data unit:	MWh
Description:	Electricity delivered to the grid by the proposed project
Source of data to be used:	Directly measured
Value of data applied for the purpose of calculating expected emission reductions in section B.5	124,660
Description of measurement methods and procedures to be applied:	The metering equipment will be properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448 －2000). The meter is installed at the transformer station, which measures the total electricity supplied to the grid. Through the electronic multifunctional electricity meter (accuracy degree is 0.2S, bidirectional), the electricity supplied to the grid is measured. The readings of electricity meter will be continuously measured and monthly recorded. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.
QA/QC procedures to be applied:	The electricity output from each turbine will be monitored and recorded. The project operator is responsible for recording this set of data. Electricity sales invoices will also be obtained for double check.
Any comment:	Electricity supplied by the project activity to the grid. Double check by receipt of sales.

B.7.2. Description of the monitoring plan:

This monitoring plan will be implemented by professional staff authorized by the owner of the proposed project. The management structure is illustrated as follows:



B.8. Date of completion of the application of the baseline study and monitoring methodology and the name of the responsible person(s)/entity(ies)

The baseline study of the proposed project was completed on 14/11/2007. It has received abundant and valuable guidance from Global Climate Change Institute (GCCI) of Tsinghua University.

The persons involved in baseline study are listed as follows:

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(Not the project participants listed in Annex 1)

SECTION C. Duration of the project activity / Crediting period

C.1 Duration of the project activity:

C.1.1. Starting date of the project activity:

10/04/2006. (the date of construction permission)

C.1.2. Expected operational lifetime of the project activity:

20 years.

C.2 Choice of the crediting period and related information:

C.2.1. Renewable crediting period

C.2.1.1. Starting date of the first crediting period:

01/03/2008 or the date of registration whichever is later..

C.2.1.2. Length of the first crediting period:

7 years.

C.2.2. Fixed crediting period:

C.2.2.1. Starting date:

N/A

C.2.2.2. Length:

N/A.

SECTION D. Environmental impacts

D.1. Documentation on the analysis of the environmental impacts, including transboundary impacts:

The Environmental Assessment Report of the proposed project has been approved by the Environmental Protection Administration of Inner Mongolia.

The EIA shows that: being located in the high-latitude grassland, the air quality of the proposed project site is relatively well, which meets the Class I of China Ambient Air Quality Standard (GB 3095-1996); the environmental noise meets Class II of China Environmental Noise Standard in Urban Area (GB 3096-93); and the quality of groundwater meets the Class III of China Quality Standard for Ground Water (GB/T 14848-93), which can be used as drinking water resource.

The EIA shows that the environmental impact will occur during the construction period and operation period.

The waste water released during the construction period will be mainly living waste water. A leakage proof cesspool will be built in the living area for construction staff. The waste water will be treated in the cesspool and then be reused as irrigation water, and will thus have little impact on the surface water. The solid living waste will be transported to landfills. Dumping waste in the proposed project site is forbidden to minimize environmental impacts. The impacts on the sound environment and air quality are short-term during the construction period. After the completion of the construction, the impacts will disappear naturally. The occupation of ground will destroy some surface vegetation during the construction period, but the vegetation destroyed by temporary ground occupation will be recovered after the completion of the construction. The impacts on local ecology system will then be minimized.

There will be no air pollutant emissions during the operation period of the proposed project. There will be a little amount of living waste water released during the operation period of the proposed project, which will be treated in the leakage proof cesspool and then be reused in farming irrigation. Therefore, the waste water will have little impact on the surface water. The noise source during the operation period of the proposed project will be mainly from the running of the wind power unit, which will have no impacts on the area 500m away. Since the site of the proposed project is on the top of mountain, there will be no noise impacts on local environment. The electromagnetism of wind power project operation will be relatively low, and will have little impact on the neighbourhood environment. The operation of the proposed project will have slight impacts on birds.

In conclusion, being as a typical type of clean renewable energy, the proposed project has no significant impacts on local environment and will greatly contribute to achievement of sustainable development objective and promote local environmental protection.

D.2. If environmental impacts are considered significant by the project participants or the host Party, please provide conclusions and all references to support documentation of an environmental impact assessment undertaken in accordance with the procedures as required by the host Party:

Not applicable, since the construction and operation of the proposed project have no significant environmental impacts.

SECTION E. Stakeholders' comments

E.1. Brief description how comments by local stakeholders have been invited and compiled:

On July, 26th, 2006, under the support of the local municipal government, the project owner successfully held a stakeholder meeting in Wengniute County. Stakeholder representatives participated the meeting, respectively from the local Government, local Environmental Protection Bureau, local Development and Reform Bureau, local Wind Power Development Office, local Agricultural Power Bureau, local National Land Bureau, and Yangshugoumen Village in Wengniute County.

E.2. Summary of the comments received:

Every stakeholder representative expressed the comments for the proposed project. No opposite comment was received. The summary of the comments is as follows:

Comments from the local government: The proposed project has been approved by the Development and Reform Commission of Inner Mongolia and Environmental Protection Administration of Inner Mongolia, which shows that the construction and operation of the proposed project will have little impacts on the local environment. The proposed project is one of the largest investment projects in local area. The local government highly supports the proposed project, and expects the increase of local financial incoming and new employment opportunity through the implementation of the proposed project.

Comments from the local Agricultural Power Bureau: Wind Power is renewable energy and is helpful in diversifying power mix of local power grid. Furthermore, wind power takes up a small share in the local power grid and will not have much impact.

Comments from villager representatives: The proposed project site is located on a mesa. There are no residents and only a few croplands around the proposed project. Therefore, there is no issue on noise disturbance and residents movement. Moreover, the project owner has made compensation for the land occupied by the proposed project. The local residents also benefit from the employment opportunities for construction and operation of the proposed project.

E.3. Report on how due account was taken of any comments received:

Since there is no negative comment received, it's no need to make adjustment on design, construction and operation of the proposed project.

Annex 1

CONTACT INFORMATION ON PARTICIPANTS IN THE PROJECT ACTIVITY

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Annex 2

INFORMATION REGARDING PUBLIC FUNDING

No public funding from Annex I countries is involved in the proposed project.

Annex 3

BASELINE INFORMATION

The Annex 3 provides the basic data and calculation for determining baseline and IRR.

Table A-1～A-3 provides the Thermal power electricity generation in Northeast Power Grid in the baseline scenario from 2003 to 2005, in which the main data sources come from China electric Power Yearbook 2004, 2005 and 2006.

Table A-1 Thermal power electricity generation in Northeast Power Grid in 2003

Province	Generating capacity (MWh)	Rate of electricity consumption (%)	Power supply (MWh)
Liaoning	79751000	7.17	74,032,853.30
Jilin	29739000	7.32	27,562,105.20
Heilongjiang	48493000	8.48	44,380,793.60
Total (MWh)			**145,975,752.10**

《China Electric Power Yearbook 2004》P709，P670

Table A-2 Thermal power electricity generation in Northeast Power Grid in 2004

Province	Generating capacity (MWh)	Rate of electricity consumption (%)	Power supply (MWh)
Liaoning	84543000	7.21	78,447,449.70
Jilin	33242000	7.68	30,689,014.40
Heilongjiang	53482000	7.84	49,289,011.20
Total (MWh)			**158,425,475.30**

《China Electric Power Yearbook 2005》P472，P474

Table A-3 Thermal power electricity generation in Northeast Power Grid in 2005

Province	Generating capacity (MWh)	Rate of electricity consumption (%)	Power supply (MWh)
Liaoning	83697000	7.03	77,813,100.90
Jilin	35294000	6.59	32,968,125.40
Heilongjiang	58000000	7.96	53,383,200.00
Total (MWh)			**164,164,426.30**

《China Electric Power Yearbook 2006》

Key parameters for the emission factors calculation

The key parameters in OM and BM calculation include the net caloric values (*NCVs*), oxidation factors (*OXIDs*), and CO_2 emission factor per unit of energy ($EF_{co2}s$) of various types of fuels, and power supply efficiency of various power generation technologies.

Table A-4 NCVs, OXIDs, and EFco2s of various types of fuels

Fuel	NCV	EF_{co2} (tc/TJ)	OXID
Coal	20908 kJ/kg	25.80	1
Washed coal	26344 kJ/kg	25.80	1
Other Washed Coal [1]	8363 kJ/kg	25.80	1
Coke	28435 kJ/kg	29.20	1
Crude oil	41816 kJ/kg	20.00	1
Gasoline	43070 kJ/kg	18.90	1
Kerosene	43070 kJ/kg	19.60	1
Diesel	42652 kJ/kg	20.20	1
Fuel oil	41816 kJ/kg	21.10	1
Other petroleum products[2]	38369 kJ/kg	20.00	1
Natural gas	38931 kJ/m^3	15.30	1
Coke oven gas[11]	16726 kJ/m^3	12.10	1
Other gas[12]	5227 kJ/m^3	12.10	1
LPG	50179 kJ/kg	17.20	1
Refinery gas	46055 kJ/kg	15.70	1

Data sources:

NCVs are from China Energy Statistical Yearbook 2006, P287.

$EFco_2$ are from 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 1, table 1-3.

OXID are from 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 1, table 1-4.

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[1] Other washed coal includes middlings and slimes. The NCV value of middlings is adopted here, which is conservative because the NCV value of slimes is higher than that of middlings.

[2] The NCV value of other petroleum products are not provided in China Energy Statistical Yearbooks. This Annex calculates it as 38369 kJ/kg, i.e., 1.3108 tce/t, on the basis of Energy Balance Sheets (physical quantity) and conversion factor against SCE

[11] The NCV value here adopts the lower limit of the NCV value range, i.e., 16726-17981 kJ/m^3, for coke oven gas provided in China Energy Statistical Yearbook 2005, P 365.

[12] The NCV value here adopts the lowest NCV value among those for gas by furnace, gas by heavy oil catalytic cracking, gas by heavy oil catalytic thermal cracking, gas by pressure gasification, and water coal gas, which are provided in China Energy Statistical Yearbook 2005, P 365.

Table A-5 Caculation of emission factor of advanced electricity generation technology

	variable	efficiency of electricity transmission	Emission Factor of fuel	Carbon oxidation rate	Emission Factor of power plant
		A	B	C	D=3.6/A/100*B*C*44/12
Coal-fired power plant	EF*coal,adv*	35.82%	25.8	1	0.9508
Gas-fired power plant	EF*gas,adv*	47.67%	15.3	1	0.4237
Oil-fired power plant	EF*oil,adv*	47.67%	21.2	1	0.5843

1. OM emission factor calculation of NEPG (Northeast Power Grid).

According to the ACM0002 methodology, the Simple method OM was used to calculate the OM emission factors of the years 2003, 2004 and 2005, and then weighted average emission coefficient was calculated and selected as the $EF_{OM,y}$ for primary fuel input for thermal power supply to the North East China grid.

The power data and processes for the calculation of the EF_{OM}，y in the Northeast China grid were shown in tables A-6 ～ A-9. The detailed calculation formulas are described in the section B6.

Table A-6 The fuel consumption and total emissions of Northeast Power Grid in 2003

Fuel	Unit	Liaoning	Jilin	Heilongjiang	Sub-Total	Carbon content (tc/TJ)	OXID (%)	NCV (MJ/t,m^3,tce)	CO_2 emissions (tCO_2e)
		A	B	C	D=A+B+C	E	F	G	H=D*E*F*G*44/12/100
Raw coal	Mt	35.5651	20.0666	27.6362	83.2679	25.8	100	20908	164695313.0
Clean coal	Mt	0.7083	0	0.03	0.7383	25.8	100	26344	1839948.7
Other washed coal	Mt	6.1704	0.159	0.5341	6.8635	25.8	100	8363	5429988.0
Coke oven gas	Billion m^3	0.166	0	0	0.166	12.1	100	16726	123184.8
Other gas	Billion m^3	0.531	0	0	0.531	12.1	100	5227	123141.3
Crude oil	Mt	0.0339	0	0	0.0339	20	100	41816	103954.6
Diesel	Mt	0.0032	0.0034	0	0.0066	20.2	100	42652	20850.0
Fuel oil	Mt	0.1487	0.007	0.0432	0.1989	21.1	100	41816	643474.2
LPG	Mt	0.0155	0	0	0.0155	17.2	100	50179	49051.6
Refinery gas	Mt	0.0403	0	0.0046	0.0449	15.7	100	46055	119040.4
Natural gas	Billion m^3	0	0.004	0.447	0.451	15.3	100	38931	984997.1
Other energy	Mtce	0.2938	0	0	0.2938	0		29271.2	0.0
Total									**174132943.7**

Data sources: China Energy Statistical Yearbook 2004

Table A-7 The fuel consumption and total emissions of Northeast Power Grid in 2004

Fuel	Unit	Liaoning	Jilin	Heilongjiang	Sub-Total	Carbon content （tc/TJ）	OXID （%）	NCV （MJ/t,m³,tce）	CO_2 emissions （tCO_2e）
		A	B	C	D＝A+B+C	E	F	G	H=D*E*F*G*44/12/100
Raw coal	Mt	41.442	23.109	30.848	95.399	25.8	100	20908	188689376.8
Clean coal	Mt	0.8475	0.0109	0.0488	0.9072	25.8	100	26344	2260871.6
Other washed coal	Mt	5.7767	0.1426	0.61	6.5293	25.8	100	8363	5165589.1
Coke oven gas	Billion m^3	0.483	0.291	0	0.774	12.1	100	16726	574367.5
Other gas	Billion m^3	5.733	0.419	0	6.152	12.1	100	5227	1426676.9
Crude oil	Mt	0	0	0	0	20	100	41816	0.0
Diesel	Mt	0.0204	0.0116	0.0024	0.0344	20.2	100	42652	108672.7
Fuel oil	Mt	0.1281	0.0178	0.0286	0.1745	21.1	100	41816	564536.2
LPG	Mt	0.0219	0	0	0.0219	17.2	100	50179	69305.2
Refinery gas	Mt	0.0979	0	0.0114	0.1093	15.7	100	46055	289779.7
Natural gas	Billion m^3	0	0.003	0.253	0.256	15.3	100	38931	559111.4
Other energy	Mtce	0.2697	0.0507	0	0.3204	0		29271.2	0
Total									**199708287.3**

Data sources: China Energy Statistical Yearbook 2005

Table A-8 The fuel consumption and total emissions of Northeast Power Grid in 2005

Fuel	Unit	Liaoning	Jilin	Heilongjiang	Sub-Total	Carbon content (tc/TJ)	OXID (%)	NCV (MJ/t,m3,tce)	CO_2 emissions (tCO2e)
		A	B	C	D＝A+B+C	E	F	G	H=D*E*F*G*44/12/100
Raw coal	Mt	43.0541	24.4613	33.8321	101.3475	25.8	100	20908	200454895.9
Clean coal	Mt	0	0	0	0	25.8	100	26344	0.0
Other washed coal	Mt	5.2474	0.1926	0.2416	5.6816	25.8	100	8363	4494939.9
Coke oven gas	Billion m^3	0.103	0.357	0.068	0.528	12.1	100	16726	391816.6
Other gas	Billion m^3	1.262	0.837	0	2.099	12.1	100	5227	486767.7
Crude oil	Mt	0.0116	0	0	0.0116	20	100	41816	35571.5
Diesel	Mt	0.0118	0.0148	0.0057	0.0323	20.2	100	42652	102038.7
Fuel oil	Mt	0.0932	0.0246	0.0155	0.1333	21.1	100	41816	431247.4
LPG	Mt	0.0012	0	0	0.0012	17.2	100	50179	3797.5
Refinery gas	Mt	0.0548	0	0.0132	0.068	15.7	100	46055	180283.8
Natural gas	Billion m^3	0	0.084	0.224	0.308	15.3	100	38931	672681.0
Other energy	Mtce	0.1618	0	0	0.1618	0	100	29271.2	0.0
Total									**207254040**

Data sources: China Energy Statistical Yearbook 2006

Table A-9 The OM factor of Northeast Power Grid

Years	Thermal generation delivered to Northeast Power Grid	The emissions from Northeast Power Grid	OM
	A	**B**	**C=B/A**
2003	145975752.1	174132943.7	1.192889512
2004	158425475.3	199708287.3	1.260581904
2005	164,164,426.30	207254040.0	1.262478386
Total	**468565653.7**	**581095271.0**	
Average OM			**1.2402**

2. Calculation of the Build Margin Emission Factor （$EF_{BM,y}$）

According to the ACM0002 methodology, the Build Margin emission factor $EF_{BM,y}$ *ex-ante* was selected to identify sample group for calculating Build Margin emission factor. Based on the description of formulas in section B6, the Build Margin emission factor is calculated to be **0.8632** tCO_2/MW・h.
The power data and processes for the calculation of the $EF_{BM,y}$ in the North East China grid were shown in tables A-10～A-14 . The detailed calculation formulas are described in the section B6.

Step 2a: calculating the respective percentages of CO_2 emissions from coal fired power generation, oil fired power generation, and gas fired power generation against total CO_2 emissions from fossil fuel fired power generation

Table A-10 Calculation of emission weight of solid fuel, liquid fuel and gas fuel in all fuel emission

Fuel	unit	Liaoning	Jilin	Heilongjiang	Total	NCV (MJ/t, km^3, tce)	Emission Factor (Tc/TJ)	Carbon oxidation rate (%)	CO_2 emission (tCO_2e)
		A	**B**	**C**	**D=A+B+C**	**H**	**I**	**J**	**K=G*H*I*J*44/12/100**
Raw coal	10^4t	4305.41	2446.13	3383.21	10134.75	20908	25.80	1	200,454,896
Clean coal	10^4t	0	0	0	0.00	26344	25.80	1	0
Other washed coal	10^4t	524.74	19.26	24.16	568.16	8363	25.80	1	4,494,940
coke	10^4t	0	0	0	0.00	28435	29.20	1	0
Sub-total									**204,949,836**
Crude oil	10^4t	1.16	0	0	1.16	41816	20.00	1	35,571
Gasoline	10^4t	0	0	0	0	43070	18.90	1	0
Kerosene	10^4t	0	0	0	0	43070	19.60	1	0
Diesel	10^4t	1.18	1.48	0.57	3.23	42652	20.20	1	102,039
Fuel oil	10^4t	9.32	2.46	1.55	13.33	41816	21.10	1	431,247
other petroleum product	10^4t	0	0	0	0	38369	20.00	1	0
Sub-total					0				**568,858**
Natural gas	10^7m^3	0	8.4	22.4	30.8	38931	15.30	1	672,681
COG	10^7m^3	10.3	35.7	6.8	52.8	16726	12.10	1	391,817
Other Gas	10^7m^3	126.2	83.7	0	209.9	5227	12.10	1	486,768
LPG	10^4t	0.12	0	0	0.12	50179	17.20	1	3,798
Refinery gas	10^4t	5.48	0	1.32	6.8	46055	15.70	1	180,284
Sub-total									**1,735,347**
Total									**207,254,040**

Data sources: *China Energy Statistical Yearbook 2006*

From above table and formulae (5),(6) and (7),the weights are as follows:
λ_{Coal} =98.89%，λ_{Oil} =0.47%，λ_{Gas} =0.84%

Step 2b: calculating the corresponding emission factor for fossil fuel fired power generation

$$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal,Adv} + \lambda_{Oil} \times EF_{Oil,Adv} + \lambda_{Gas} \times EF_{Gas,Adv} = 0.9453.$$

Step 2c: calculating the $EF_{BM,y}$ of local grid

Table A-11 Installed capacity of Northeast Power Grid in 2005

Installed capacity	unit	Liaoning	Jilin	Heilongjiang	total
Thermal Power	MW	15999	6359.4	11575.6	33934
Hydropower	MW	1403.9	3720.8	846.7	5971.4
Nuclear	MW	0	0	0	0
Wind power and other	MW	135.5	85.4	52.4	273.3
Total	MW	17538.4	10165.6	12474.7	40178.7

Data sources:《China Electric Power Yearbook2006》

Table A-12 Installed capacity of Northeast Power Grid in 1999

Installed capacity	unit	Liaoning	Jilin	Heilongjiang	total
Thermal Power	MW	12425.7	4583.1	10128.1	27136.9
Hydropower	MW	1240	3508.2	774.5	5522.7
Nuclear	MW	0	0	0	0
Wind power and other	MW	22.9	0	0	22.9
Total	MW	13688.6	8091.3	10902.6	32682.5

Data sources:《China Electric Power Yearbook2000》

Table A-13 Installed capacity of Northeast Power Grid in 1998

Installed capacity	unit	Liaoning	Jilin	Heilongjiang	total
Thermal Power	MW	12560.3	4428.6	9116	26104.9
Hydropower	MW	1223.1	3474.7	784.5	5482.3
Nuclear	MW	0	0	0	0
Wind power and other	MW	17	0	0	17
Total	MW	13800.4	7903.3	9900.5	31604.2

Data sources:《China Electric Power Yearbook1999》

Table A-14 The new installed capacity from 1998-2005 in the Northeast Power Grid

	Installed capacity in 1998	Installed capacity in 1999	Installed capacity in 2005	Addition capacity from 1998 to 2005	Addition share**(%)**
	A	B	C	D=C-A	
Thermal Power	26104.9	27136.9	33934	7829.1	91.31%
Hydropower	5482.3	5522.7	5971.4	489.1	5.70%
Nuclear	0	0	0	0	0.00%

Wind power	17	22.9	273.3	256.3	2.99%
Total（MW）	31604.2	32682.5	40178.7	8574.5	100.00%
Share of 2004 installed capacity	78.66%	81.34%	100.00%		

$EF_{BM,y}=EF_{Thermal}\times CAP_{Thermal}/CAP_{Total}=0.9453\times 91.31\%=0.8362$

where:

CAP_{Total} is the total capacity addition,

$CAP_{Thermal}$ is the fossil fuel fired capacity addition.

3. Calculation of the Baseline Emissions Factor（EF_y）

According to the baseline methodology (ACM0002), the baseline emission factor EF_y is calculated as the weighted average of the Operating Margin emission factor ($EF_{OM,y}$) and the Build Margin emission factor ($EF_{BM,y}$), as shown in table A-15:

Table A-15 Baseline Emission factor (EF_y) of the Northeast Power Grid

Calculation of the Key factors:
Operating Margin emission factor（$EF_{OM,y}$）（tCO_2/MWh）： 1.2402
Build Margin emission factor（$EF_{BM,y}$） （tCO_2/MWh）： 0.9453×9131%=0.8632
Baseline Emission factor（EF_y） （tCO_2/MWh）： 1.2404×0.75+0.8632×0.25=1.1459

Note: the latest version of ACM0002 (version 6) provides the following default weights for wind and solar projects: Operating Margin, W_{OM} = 0.75; Build Margin, W_{BM} = 0.25.

4. IRR calculation of the proposed project.

Tables B1 show the Parameters needed for calculation of IRR.

The below Formulas are used in the IRR calculation process. They are based on «Method and Parameter of economic analysis for construction project » published by National Development and Reform Commission:

a) Cash inflow= sales revenue+ fixed assets residue+ recovered liquid capital
b) Sales revenue= annual output× tariff (excl. VAT)
c) Fixed assets residual value= original fixed assets value × rate of assets residual value
d) Recovered liquid capital= liquid capital input at beginning of project operation
e) Cash outflow= capital construction investment + liquid capital+ operating cost + sales tax and extra charges + income tax

Where:
Capital construction investment = Static total investment
Liquid capital = Liquid capital input of current year;
Operating cost= annual salary per capita ×employee population × (1+ rate of welfarism) + the sum of original value of buildings and equipment × (rate of maintenance + rate of

insurance premium) + (fixed amount of material cost+ fixed amount of other costs) × installed capacity;
Sales tax and extra charges=sales revenue × rate of VAT × (rate of city construction tax + rate of additional education fee);
Income tax= 〔sales revenue- sales tax and extra charges - operating cost - original value of houses and buildings × (1- expected rate of residual value) ÷ expected depreciable life - machinery × (1- expected rate of residual value) ÷ expected depreciable life - other assets ÷ amortizing period〕 × rate of income tax;

f) Net cash flow = cash inflow - cash outflow

g) The FIRR of the proposed project is calculated with the following formula:

$$\sum_{1}^{n} \frac{annual\ net\ cash\ flow_n}{(1+FIRR)^n} = 0$$

where:
n is the project life time(including construction period and operation period);
$annual\ net\ cash\ flow_n$ is the net cash flow in NO. n year.
FIRR is the financial internal rate of return, a financial indicator of the proposed project.

h) Net cash flow with CERs income = net cash flow without CERs income +〔CERs income excl. VAT －CERs income excl. VAT × VAT rate ×(rate of city construction tax + rate of additional education fee))×（1 - rate of income tax)

i) The FIRR of the proposed project with CERs income is calculated with the following formula:

$$\sum_{1}^{n} \frac{annual\ net\ cash\ flow_n^{'}}{(1+FIRR^{'})^n} = 0$$

where:
n is the project life time;
$annual\ net\ cash\ flow_n^{'}$ is the net cash flow with CERs income in NO. n year.
$FIRR^{'}$ is the financial internal rate of return of the proposed project with CERs income.

Table B-1 Main Parameters needed for calculation of key financial indicators

No.	Item	Unit	Figure
1	Type of wind turbines	***	750
2	Number of wind turbines		67
3	Installed capacity	MW	50.25
4	Annual operation hours	Hour	2518
5	Annual generation delivered to the grid	10000 kWh	12466
6	Electricity tariff(Excl. VAT)	Yuan/kWh	0.5073
7	Static total investment	10000 Yuan	47266
8	Liquid capital	10000 Yuan	50
9	Construction period	Year	1

10	Operation lifetime	Year	20
11	Depreciable life of fixed assets	Year	12
12	Rate of residual value of fixed assets	%	4
13	Amortization period of other assets	Year	0
14	Rate of fixed assets maintenance	%	0.09%[13]
15	Rate of insurance premium of fixed assets	%	0.11%[14]
16	Employee population		20
17	Annual salary per capita	10000 Yuan	2.85
18	Rate of welfarism	%	41
19	Material cost	Yuan/kW	0
20	Other costs	Yuan/ kW	40
21	Rate of VAT	%	8.5
22	Rate of city construction tax	%	5
23	Rate of additional education fee	%	3
24	Rate of income tax	%	33
25	**CERs**	Ton	142848
26	**CERs Unit price**	ERU/Ton	10.21
27	**CERs income (incl. VAT)**	10000 Yuan	1469
28	**CERs income (excl. VAT)**	10000 Yuan	1354
29	Exchange rate	ERU:RMB	10.074

Date source: items (1-6)and (8-24) are from the feasibility study report of the proposed project. The source of item 7 see page 11 of the PDD. Item 29 is according to the exchange rate on 25th Jan. 2007.

[13] The rate of fixed Assets maintenance is 0.09% in years 2 and 3, 1.34% from year 4 to year 11, and1.76% from the year 12 to the year 21.
[14] The insurance premium rate is 0.11% in years 2 and 3, and 0.405% in the years after.

Cash Flow Table (total investment)

Unit：10000 Yuan RMB

No.	Item	Total	Construction Period	Operation Period								
			1	2	3	4	5	6	7	8	9	10
	（10000kW.h）Annual output	249320	0	12466	12466	12466	12466	12466	12466	12466	12466	12466
	Tariff (Yuan/kWh, excl. VAT)	***	0.0000	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073
	Rate of fixed assets maintenance and insurance premium (%)[15]	***	0.000	0.190	0.190	1.745	1.745	1.745	1.745	1.745	1.745	1.745
	Rate of income tax (%)	***	0	33	33	33	33	33	33	33	33	33
1	Cash inflow	128421	0	6324	6324	6324	6324	6324	6324	6324	6324	6324
1.1	Sales revenue	126480	0	6324	6324	6324	6324	6324	6324	6324	6324	6324
1.2	Fixed assets residual value	1891	0	0	0	0	0	0	0	0	0	0
1.3	Recovered liquid capital	50	0	0	0	0	0	0	0	0	0	0
2	Cash outflow	89785	47266	1165	1115	1607	1607	1607	1607	1607	1607	1607
2.1	Construction investment	47266	47266									
2.2	Liquid capital	50		50								
2.3	Operating cost	22579		368	368	1103	1103	1103	1103	1103	1103	1103
2.4	Sales tax & extra charges	860	0	43	43	43	43	43	43	43	43	43
2.5	Income tax	19030		703	703	461	461	461	461	461	461	461
3	Net cash flow (1-2)	38636	-47266	5159	5209	4717	4717	4717	4717	4717	4717	4717
4	Accumulative total of net cash flow	***	-47266	-42107	-36897	-32180	-27464	-22747	-18030	-13313	-8596	-3879
5	Net cash flow with CERs income	55937	-47266	6024	6074	5582	5582	5582	5582	5582	5582	5582

[15] The insurance premium rate is merged into the rate of fixed assets maintenance.

No.	Item	Operation Period										
		11	12	13	14	15	16	17	18	19	20	21
	（10000kW.h）Annual output	12466	12466	12466	12466	12466	12466	12466	12466	12466	12466	12466
	Tariff (Yuan/kWh, excl. VAT)	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073	0.5073
	Rate of fixed assets maintenance and insurance premium (%)	1.745	2.165	2.165	2.165	2.165	2.165	2.165	2.165	2.165	2.165	2.165
	Rate of income tax (%)	33	33	33	33	33	33	33	33	33	33	33
1	Cash inflow	6324	6324	6324	6324	6324	6324	6324	6324	6324	6324	8265
1.1	Sales revenue	6324	6324	6324	6324	6324	6324	6324	6324	6324	6324	6324
1.2	Fixed assets residual value	0	0	0	0	0	0	0	0	0	0	1891
1.3	Recovered liquid capital	0	0	0	0	0	0	0	0	0	0	50
2	Cash outflow	1607	1740	1740	2988	2988	2988	2988	2988	2988	2988	2988
2.1	Construction investment											
2.2	Liquid capital											
2.3	Operating cost	1103	1302	1302	1302	1302	1302	1302	1302	1302	1302	1302
2.4	Sales tax & extra charges	43	43	43	43	43	43	43	43	43	43	43
2.5	Income tax	461	395	395	1643	1643	1643	1643	1643	1643	1643	1643
3	Net cash flow (1-2)	4717	4584	4584	3336	3336	3336	3336	3336	3336	3336	5277
4	Accumulative total of net cash flow	838	5422	10006	13342	16679	20015	23351	26687	30023	33359	38636
5	Net cash flow with CERs income	5582	5449	5449	4201	4201	4201	4201	4201	4201	4201	6142

FIRR without CERs income	**7.07%**
FIRR with CERs income	**9.74%**

Annex 4

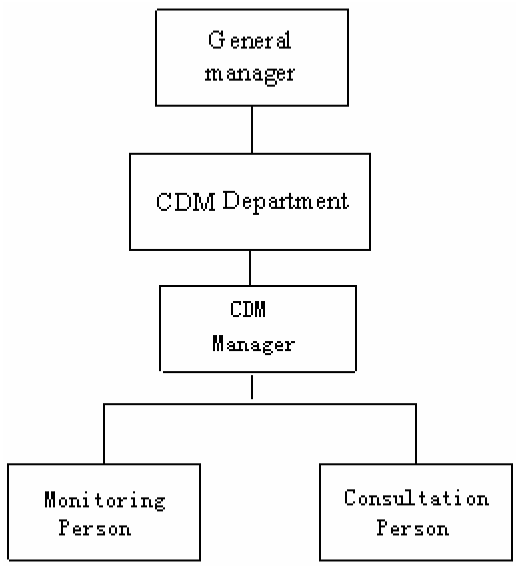
MONITORING PLAN

Monitoring plan is a division and schedule of a series of monitoring tasks. Monitoring tasks must be implemented according to the monitoring plan in order to ensure that the real, measurable and long-term greenhouse gas (GHG) emission reduction for the proposed project is monitored and reported.

1. Management structure and staff for implementation of monitoring plan

The Management Group of the proposed project must maintain credible, transparent, and adequate data estimation, measurement, collection, and tracking systems to maintain the information required for an audit of an emission reduction project.

The Management Group consists of professional staff authorized by the owner of the proposed project, i.e. Chifeng Xinsheng Wind Power Co. Ltd. The management structure is illustrated as follows:



The Management Group is responsible for the CDM management of the proposed project, which specifically includes selecting a CDM consultant company, providing support to the selected consultant company, assisting the selected consultant company in selecting CER buyer and determine CER price, assisting the selected consultant company in applying for China DNA approval, providing support in the due diligence of the selected buyer, providing support in the CDM validation and registration by a selected DOE, and providing support in the verification by a selected DOE.

The General Manager is in general control and makes key decisions. The CDM Department is responsible for specific tasks in monitoring and execution of decisions. Specifically, the CDM Manager is responsible for the daily operation of the CDM Department, and contact with DOE and EB to support their work in validation, registration, verification, and certification. The procedure is illustrated in Section 5 of this monitoring plan. He is also responsible for Monitoring Data adjustment and settlement of Data uncertainties, together with local electric power company, and/or DOE. The procedure is illustrated in Section 3 of this monitoring plan.

The Monitoring Person is responsible for reading and calibration of the meter, recording of the readings, and reporting of readings to local electric power company and/or DOE. The procedures for reading, reporting, and calibration are illustrated in Section 2 and Section 4 of this monitoring plan, respectively.

The Consultation Person is responsible for selecting a CDM consultant company and providing support to its work in developing PDD, selecting CER buyer, and applying for DNA approval.

The Management Group has all received sufficient training in terms of monitoring and verification. They have received general training on wind power project operation organized the project owner, including reading and calibration of the meter, recording of the readings, adjustment of readings, and reporting of readings. On the other hand, they have received CDM training organized by China Fulin Windpower Development Corporation, including validation, registration and verification. When necessary, the CDM Manager is responsible for organizing or attending trainings on Monitoring and Verification. The procedure is as follows:
(1) Investigating whether there is need for trainings, and if so, the content of trainings.
(2) If so, obtaining the approval of General Manager.
(3) Checking if there are any trainings on Monitoring and Verification to be organized by other organizations. If so, attending them.
(4) If not, organizing trainings themselves.

2. Data and Approach for Monitoring
The Data to be monitored include:

ID number *(Please use numbers to ease cross-referencing to table D.3)*	Data variable	Source of data	Data unit	Measured (m), calculated (c), estimated (e),	Recording frequency	Proportion of data to be monitored	How will the data be archived? (electronic/ paper)	For how long is archived data kept?	Comment
1.EG_y	Electricity supplied to the grid by the project	ammeter	MWh	*m*	continuously measured and monthly recorded	100%	electronic	During the crediting period and two years after	Electricity supplied by the project activity to the grid. rechecked by receipt of sales.

Electricity supplied to the grid by the project will be monitored through the metering equipment at the transformer station. The data can also be monitored and recorded at the on-site control centre using a computer system. The meter will have the capability to be read remotely through a communication line.
The specific steps to monitoring are listed below:

● The project owner reads the meter and records data on the same day of every month (which day to be determined).

●The project owner provides electricity sales invoice to Liaoning Electric Power Company.

● The project owner provides the meter's data readings to DOE for verification.
The meter reading will be readily accessible for DOE. Calibration test records will be maintained for verification.

3. Calibration of Meters & Metering
An agreement should be signed between the project owner and Liaoning Electric Power Company that defines the metering arrangements and the required quality control procedures to ensure accuracy.

- The metering equipment will be properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448 －2000).The metering equipment will be checked by the project owner and Liaoning Electric Power Company before operation.
- The meter is installed at the transformer station, which measures the total electricity supplied to the grid Through the electronic multifunctional electricity meter (accuracy degree is 0.2S, bidirectional), the electricity supplied to the grid is measured.
- The verification of electric energy meter should be periodically carried out according to relevant national electric industry standards or regulations. After verification, the meter should be sealed. The meter shall be jointly inspected and sealed on behalf of the parties concerned and shall not be accessible by either party except in the presence of the other party or its accredited representatives.
- The meter installed shall be tested by the qualified metrical organization co-authorized by the Northeast Power Grid and the project owner within 10 days after:

1) The detection of a difference larger than the allowable error in the reading of the meter, when considering the reactive loss of electrical wire,
2) The repair of all or part of the meter caused by the failure of one or more parts to operate in accordance with the specifications,
3) If any errors are detected, the party owning the meter shall repair, recalibrate or replace the meter and give the other party sufficient notice to allow a representative to attend during any corrective activity.

- Should reading of the meter be inaccurate by more than the allowable error, or otherwise functioned improperly, the electricity supplied to the grid by the proposed project shall be determined by:

1) First, by reading the self-carried meters of wind turbines, unless a test by either party reveals they are inaccurate;
2) if the self-carried meters of wind turbines are not with acceptable limits of accuracy or are otherwise performing improperly, the project owner and the Liaoning Electric Power Company shall jointly prepare an estimate of the correct reading; and
3) If the project owner and the Liaoning Electric Power Company fail to agree the estimate of the correct reading, then the matter will be referred for arbitration according to agreed procedures.

- The electricity recorded by the meter will suffice for the purpose of billing and emission reduction verification as long as the error in the meter is within the permissible limits.

Calibration is carried out by the Liaoning Electric Power Company with the records being provided to the project owner, and these records will be maintained by the project owner.

4. Data Management System
This provides information on record keeping of the data collected during monitoring. Record keeping is the most important exercise in relation to the monitoring process. Below follows an outline of how project related records will be managed.
Overall responsibility for monitoring greenhouse gas emissions reductions will rest with the CDM Department. The CDM manual sets out the procedures for tracking information from the primary source to data calculations, in paper format. If data and information are from internet, the website must be provided. Moreover, the credibility and reliability of those data and information from internet must be confirmed.

Physical documentation will be collated in a central place. In order to facilitate auditor's reference, monitoring results will be indexed. All paper-based information will be stored by the project owner.
The following table below outlines the key documents relevant to monitoring and verification.

Table List of the key documents relevant to monitoring and verification

I.D.No.	Document Title	Main Content	Source
F-1	PDD, including the electronic spreadsheets and supporting documentation(assumptions, estimations, measurement, etc)	Calculation procedure of emission reduction and monitoring items	The project owner
F-2	The report on monitoring and checking of electricity supplied to the grid	Record based on monthly meter reading and electricity sale receipts	The project owner
F-3	Report on maintenance and calibration of metering equipment	Reasons for maintenance and calibration and the precision after maintenance and calibration	The project owner

5. Verification and Monitoring Results

The verification of the monitoring results of the project is a mandatory process required for all CDM projects. The main objective of the verification is to independently verify that the project has achieved the emission reductions as reported and projected in the PDD. It is expected that the verification will be done annually.

The responsibilities for verification of the projects are as follows:

◆ Sign a verification service agreement with specific DOE and agree to a time framework set by the EB for carrying out verification activities while taking into account the buyer's schedule. The project owner will make the arrangements for the verification and will prepare for the audit and verification process to the best of its abilities.

◆ The project owner will facilitate the verification through providing the DOE with all required necessary information, before, during and, in the event of queries, after the verification.

◆ The project owner will fully cooperate with the DOE and instruct its staff and management to be available for interviews and respond honestly to all questions from the DOE.

◆ DOE must be an Accredited Entity with a proven track record in environmental auditing and verification, experience with CDM projects and work in developing countries. The DOE should be accredited by the CDM EB. If the project owner deems that requirements of DOE go beyond the scope of verification, they should determine whether the requirements of DOE are reasonable. If considered unreasonable, a rejection letter in a written format should be provided to the DOE with justifiable reasons. If the project owner and the DOE cannot reach an agreement, the matter will be submitted to EB or UNFCCC for arbitration.